

Product Design and Innovation

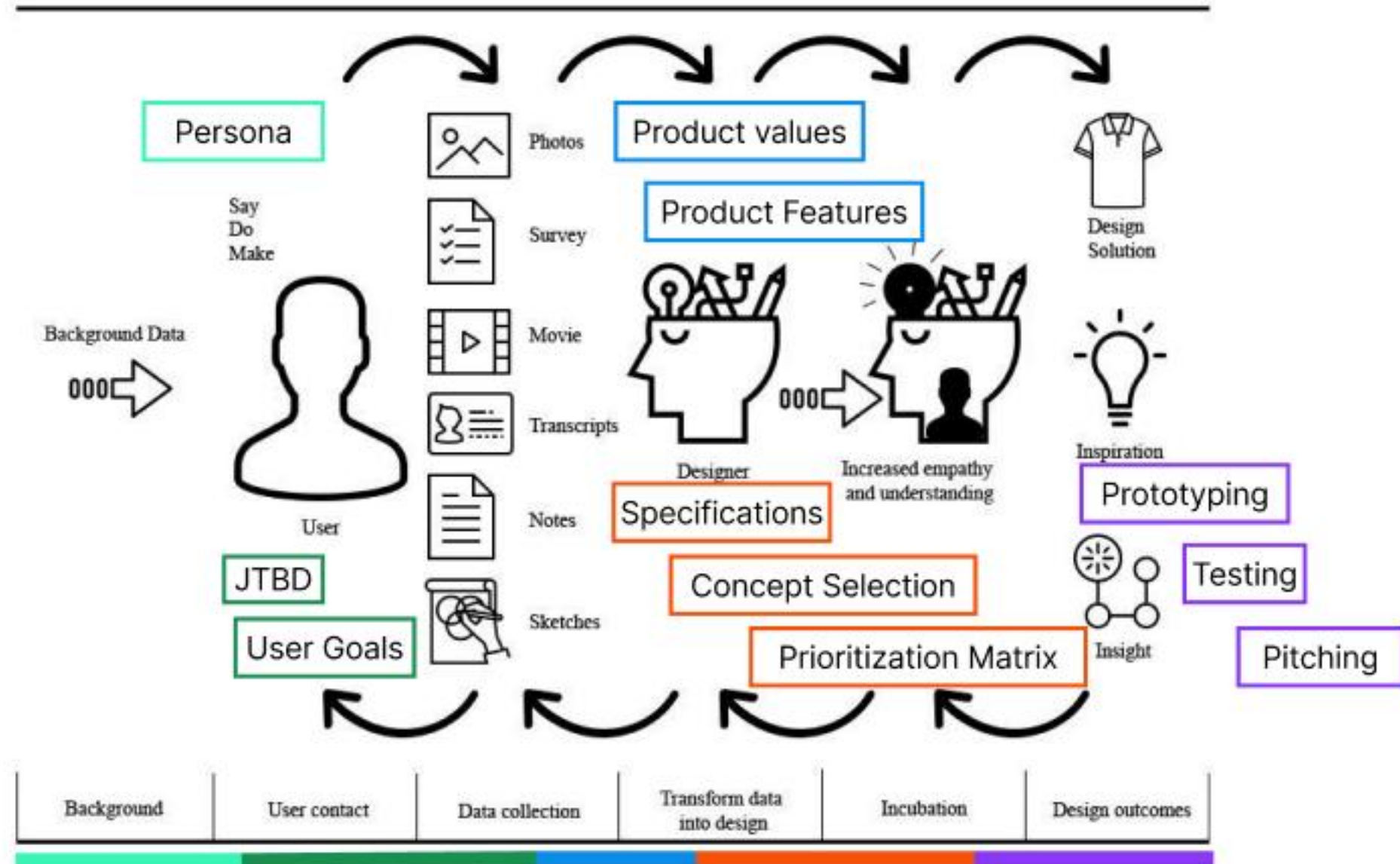


CDE2300 pdi



@CDE2300 Product Design and Innovation

Overview of Design Process

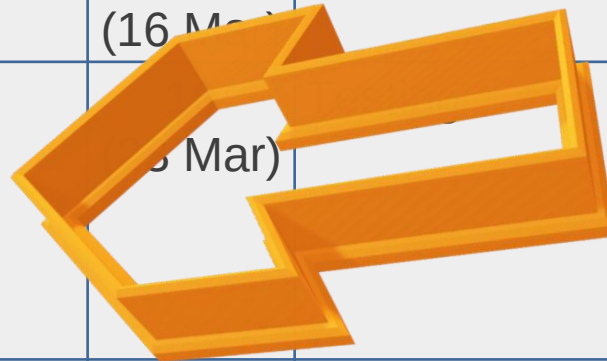


Course Overview

Week	Titles	Week	Titles
1 (12 Jan)	Innovation-driven economy, Group formation, and Brief intro	7 (2 Mar)	Concept Generation, Prioritization Map, Concept Selection, and Specifications
2 (19 Jan)	Experience economy, JTBD, Job Story and Job Statement	8 (9 Mar)	Prototyping and Prototyping Plan
3 (26 Jan)	Empathy Map, Fieldwork, and Persona	9 (16 Mar)	Storytelling
4 (2 Feb)	Use of AI in Design, Affinity Clustering, Ideation Methods, Users' Goals, Product Values and Features, and Hands-on Exercise: Crazy 8	10 (23 Mar)	Testing
5 (9 Feb)	Mid Term Review	11 (30 Mar)	Group consultation
6 (16 Feb)	No class!	12 (6 Apr)	Group consultation
23 Feb to 27 Feb	Recess Week (No Class)	13 (13 Apr)	End Term Review (Show and Tell) (2pm to 6pm)

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Assessment

Individual

- **Peer Review + Assessment (TAs) (5%)**
 - 2.5% (Mid Term), 2.5% (End Term)
 - Communication
 - Collaboration
 - Responsibility & Reliability
 - Proactive Problem Solving
 - Constructive Feedback
- **Design Reflection (due on Reading Week) (5%)**
- **Assignments (3 Homework Assignments) (5%)**

Group

- **End Term Deliverables (73%)**
- **Mid Term Deliverables (12%)**

Assessment

Mid Term Deliverables (12%)

- JTBD
- Summary of Fieldwork
- Affinity Clusters
- Persona(s)
- Product Values and Product Features

Each team is given 6 minutes to pitch and 4 minutes for Q&A. Random order. Rubrics out on 30 Jan (Friday).

To Prepare for your Fieldwork

Use the 5W and 1H method.

Who

are your
target
users?

What

do you
want to
find out?

Why

is it
important?
*(About the
Question)*

When

are you going
to conduct
the fieldwork

Where

will the
fieldwork
take place?

How

are you going to
reach out to them?
How are you going
to conduct the
fieldwork?

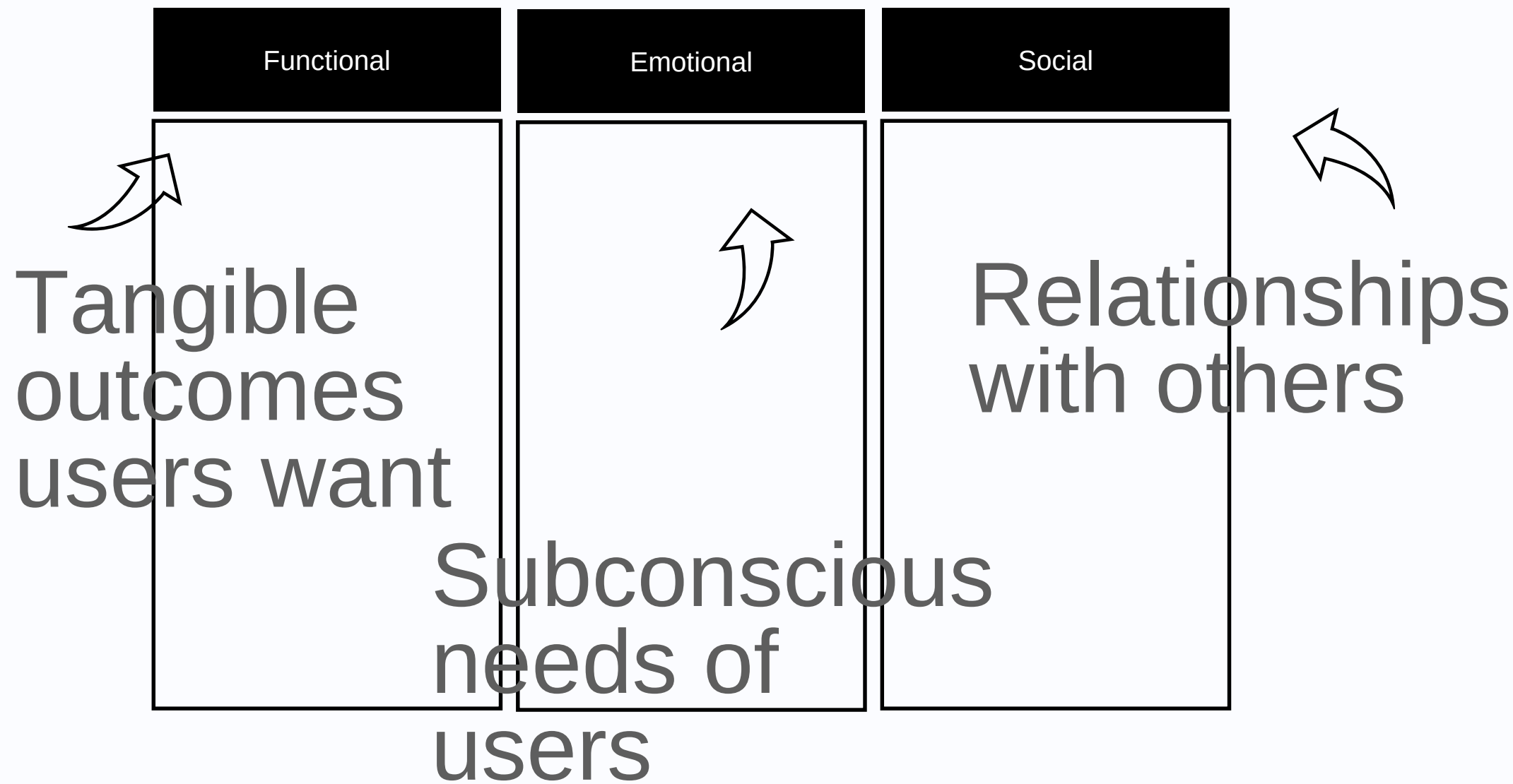
BEFORE
the Fieldwork

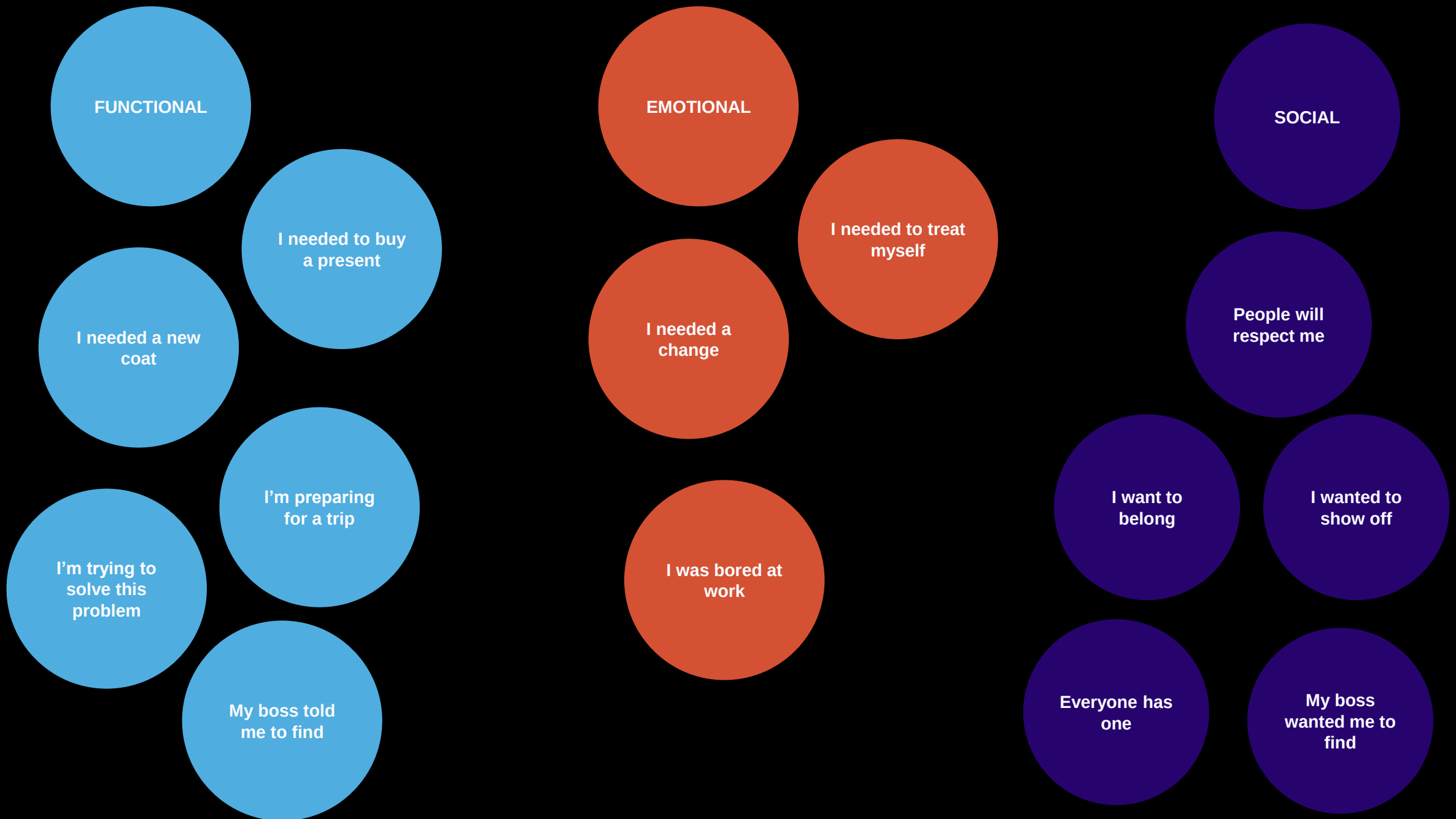


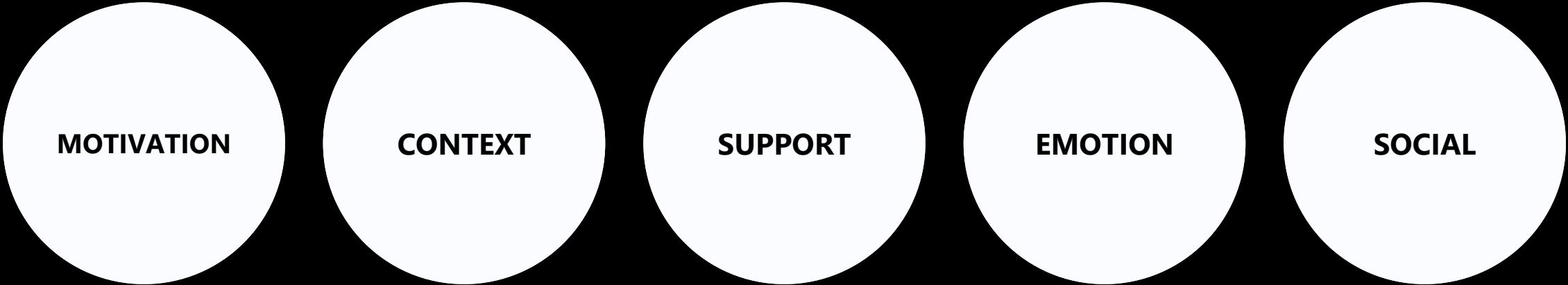
Understand customer needs (value)



JTBD Template







MOTIVATION

CONTEXT

SUPPORT

EMOTION

SOCIAL

Job Statement

WHEN I BUT HELP ME SO I



What's the context?



What's the barrier
getting in the way
today?



What is the user
trying to do?



What's the outcome of
achieving this action?

INTERVIEW your partner about a purchase

a **significant purchase** they had made in the last few months

“What have you bought recently?”

- *INTERVIEW by using JTBD framework. 20 min*
- *Write a **Job Statement**. 5 min*



CHOOSE A
PURCHASE
THAT
MATTERED IN
THE LAST 3
MONTHS

MOTIVATION

**When did you
first think
about buying
something
like this?**

Interviewee:

**Describe when you first thought about
buying something in this category.**

Interviewer:

Listen for the needs.

CONTEXT



Tell me the story about buying this product.

Interviewee:

Describe the process of buying the product in rich detail. Help us understand each step.

Interviewer:

Listen to the story. Spot emotional decisions made.

SUPPORT

**Why did you
buy this
product?**

Interviewee:

**Describe why you bought this product.
What it is for. What it does for you.**

Interviewer:

Capture obvious jobs.

EMOTION



How do you
feel about
your purchase
now?

Interviewee:

Describe how this product or service makes you feel. How well it solves your needs.

Interviewer:

Listen for future opportunities.

SOCIAL

How did you
share your
buying story
with friends
and family?

Interviewee:

Did you tell other people about your purchase? How did they react? How did that make you feel?

Interviewer:

Listen for social jobs.

Job Statement (Example)



WHEN I want to recruit new participants for CDE2300 project, **BUT I** have so many assignments to completed, **HELP ME** to automate the process of recruitment, screening, scheduling and reminder, **SO I** can quickly collect lots of data from useful participants without the enormous time spent on doing it manually.



Nerdy Nina

"The book is way better than the movie!"

#booklover

#bookaddict

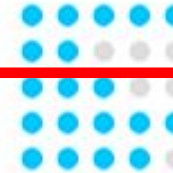
#bookloverproblems

DEMOGRAPHICS

Age: 25
Location: Sao Paulo, Brazil
Education: Software Engineer
Job: Q/A at Indie Game Company
Family: Lives with her boyfriend

TECH

Internet
Social Networks
Messaging
Games
Online Shopping



GOALS

- Discovering new books / authors to read
- Finding unique stories
- Cataloging book collection

FRUSTRATIONS

- Keeping track of different series
- Forgetting a book launch date
- Finding space for more books

READING HABITS

- Fast pace reader
- Never lends books
- Likes hardcovers and boxed collections
- Pre-order books to get them first
- Reads eBooks, but prefer physical copies
- Always finishes a book
- Loves binge reading and re-reading

FAVORITE BOOKS



American Gods
Neil Gaiman



Harry Potter
J.K. Rowling



Ready Player
One

Empathic Title

Quote

Basic Demographics

Goals & Frustrations

Habits/Preferences

A man with a shaved head, wearing a grey and white striped long-sleeved shirt, stands in a kitchen. The kitchen has wooden upper cabinets and a white tiled backsplash. On the counter in front of him are various condiments, including several bottles of sauce and a carton of tomato sauce. A dashed white circle highlights a group of small sauce bottles on the left. The man is looking slightly to his left.

Odi, 26
Surabaya

Odi creates
his own sauce

"When I make fried noodles I use tomato sauce and other sauces to make my own creation. I also use Saos Raja Rasa for fried rice."



Easy-going-about-food Iqbal

"We cook rice. We buy all other dishes. No time to cook."

Demographic

Age: 35 years old

Occupation: employee of a company

Family: married with 2 kids

Spouse: working

Experience with food

- Seldom cook at home
- Lunch: eat out near workplace
- Dinner: cook rice and buy dishes

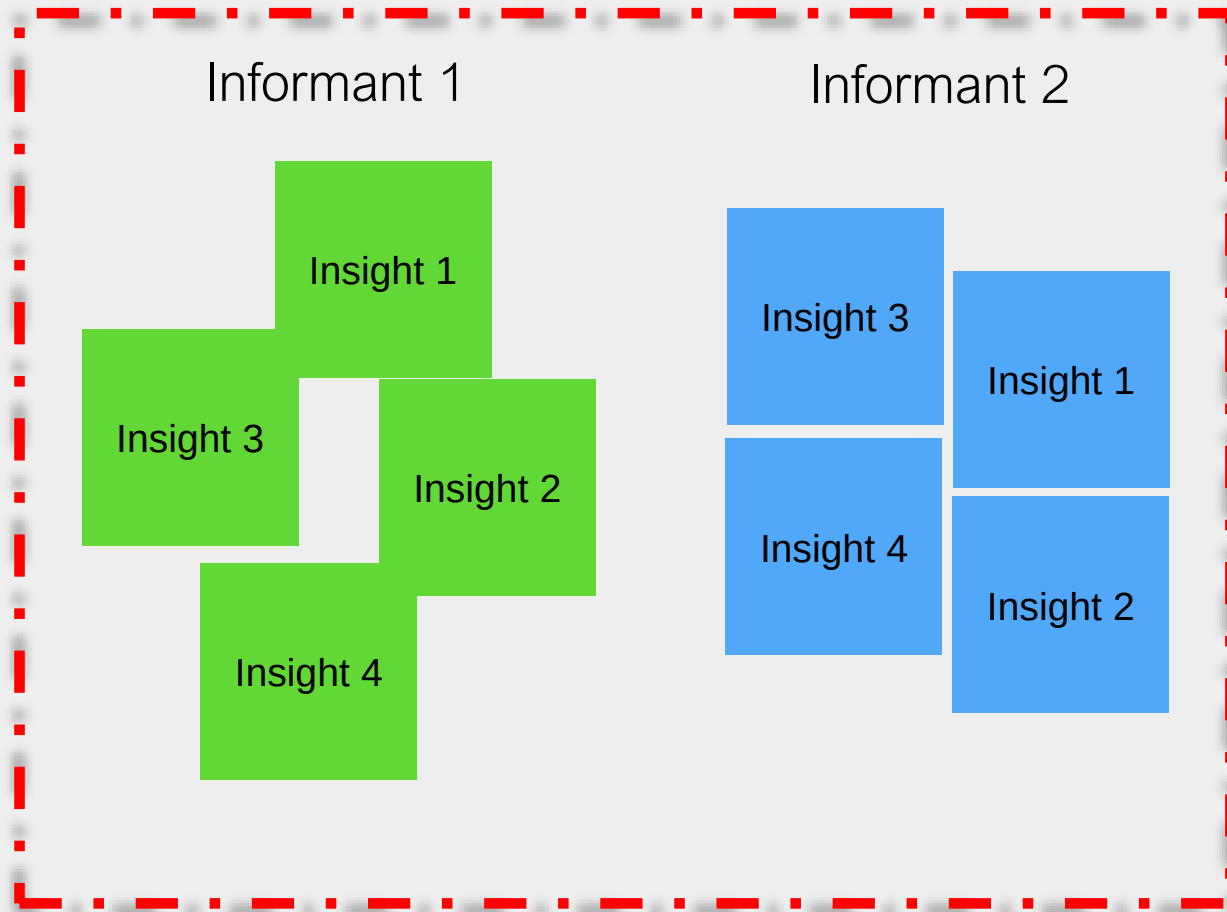
Goals

- Eat simply
- Not be overthinking about food

Frustration

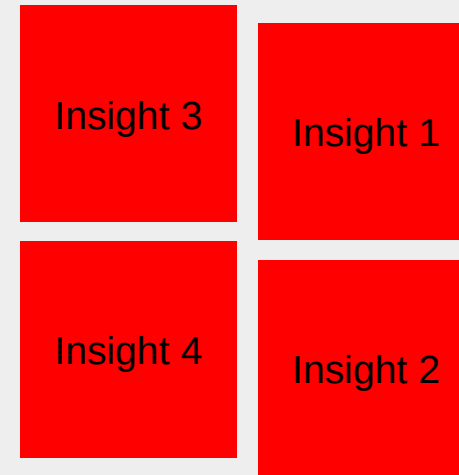
- Little time to eat with family
- Seldom be at home on weekdays

Persona 1



Persona 2

Informant 3



Persona 3

Informant 4



EMPATHY MAP

SEE? HEAR?

SAY & DO?

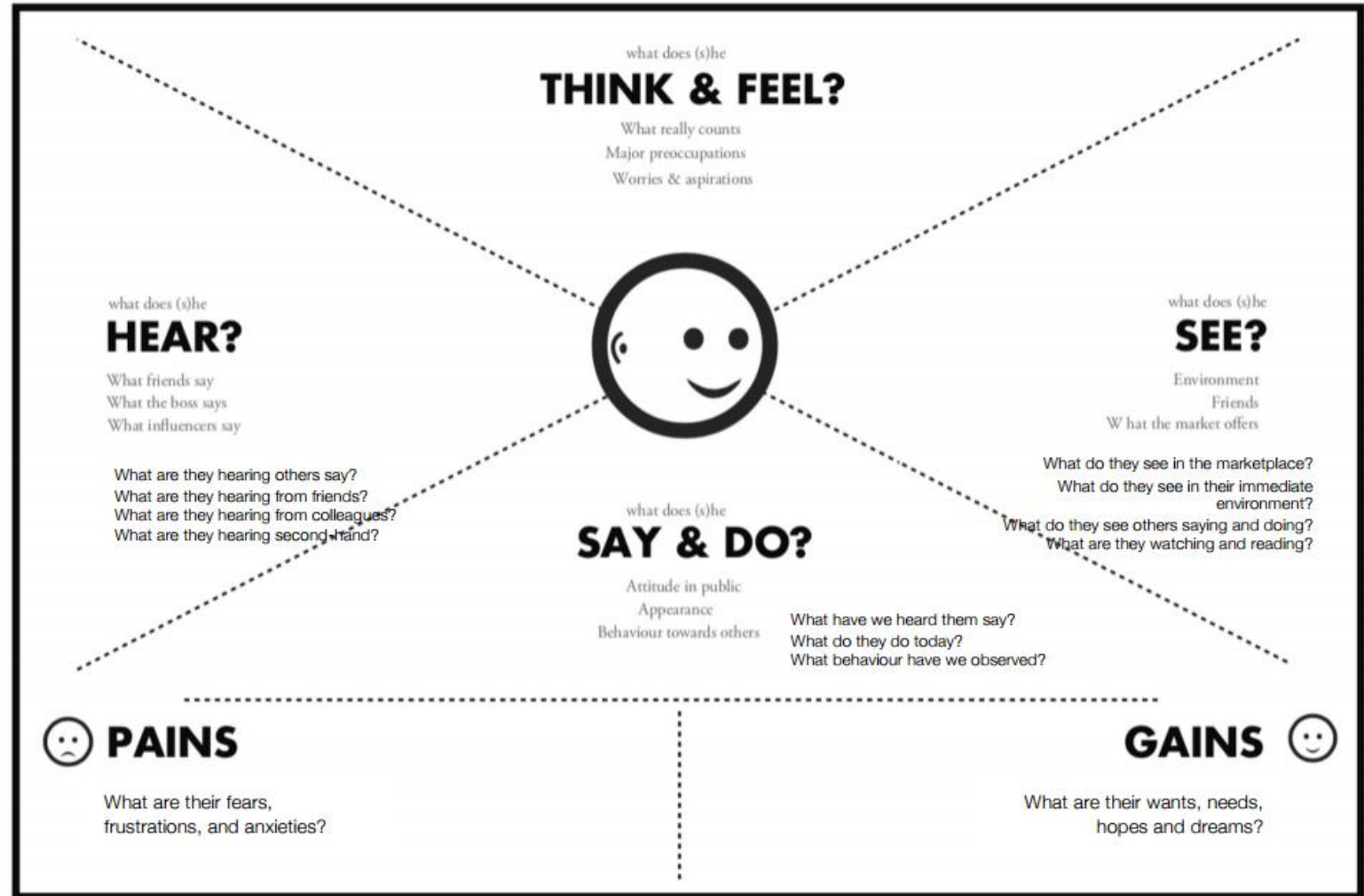
THINK & FEEL?

PAINS

- What are the pain points?
- Challenges?
- Problems or doubts
- Obstacles
- Frustrations

GAINS

- What do users hope to gain?
- What do users want to achieve?
- Goals, Needs
- What is success?
- How success is measured?



Empathy Map Canvas ©2017 David Grey

[Link to softcopy of templates for Empathy Map, JTBD and POEMS](#)



EMPATHY MAP

SEE? HEAR?

SAY & DO?

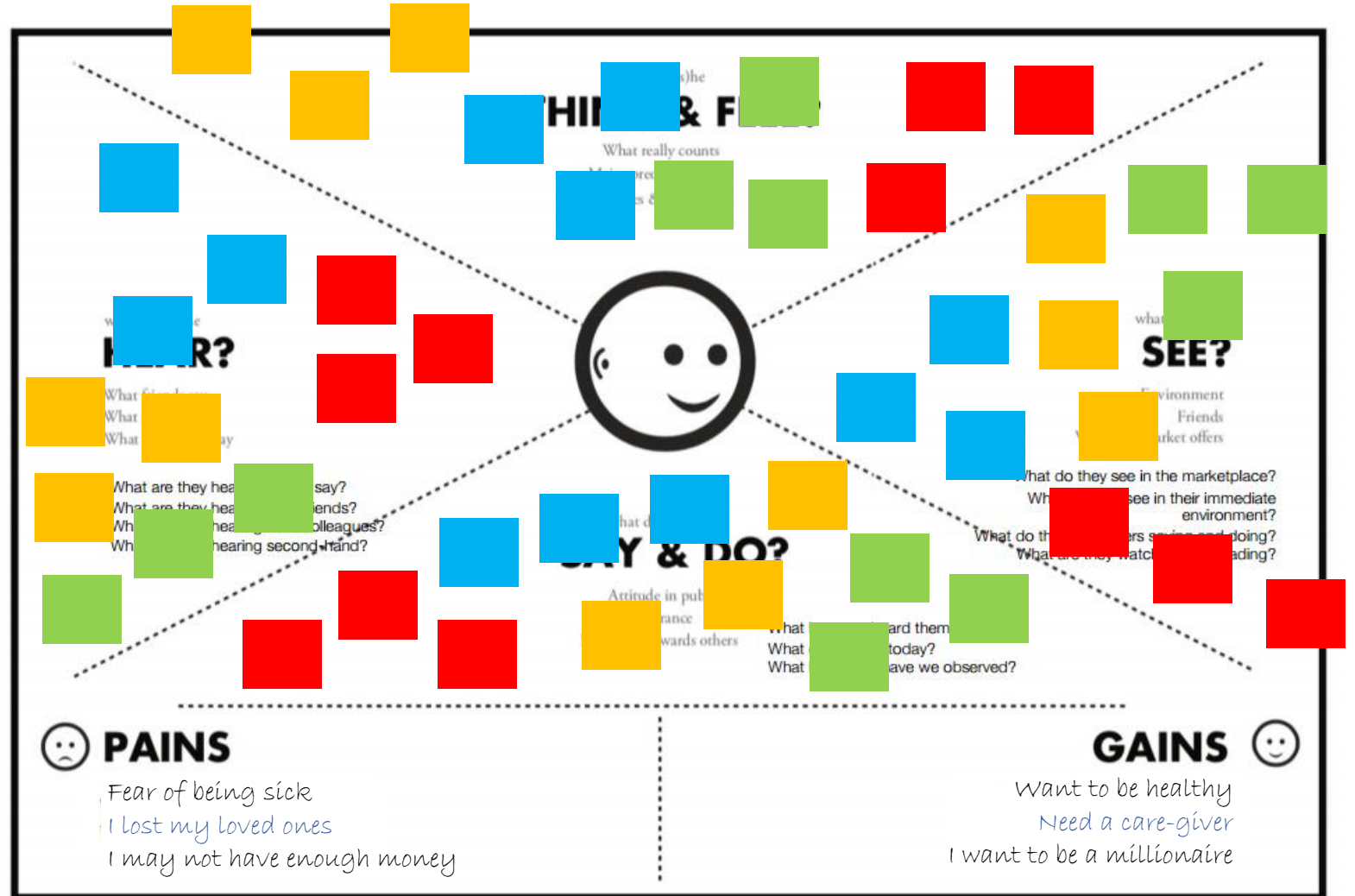
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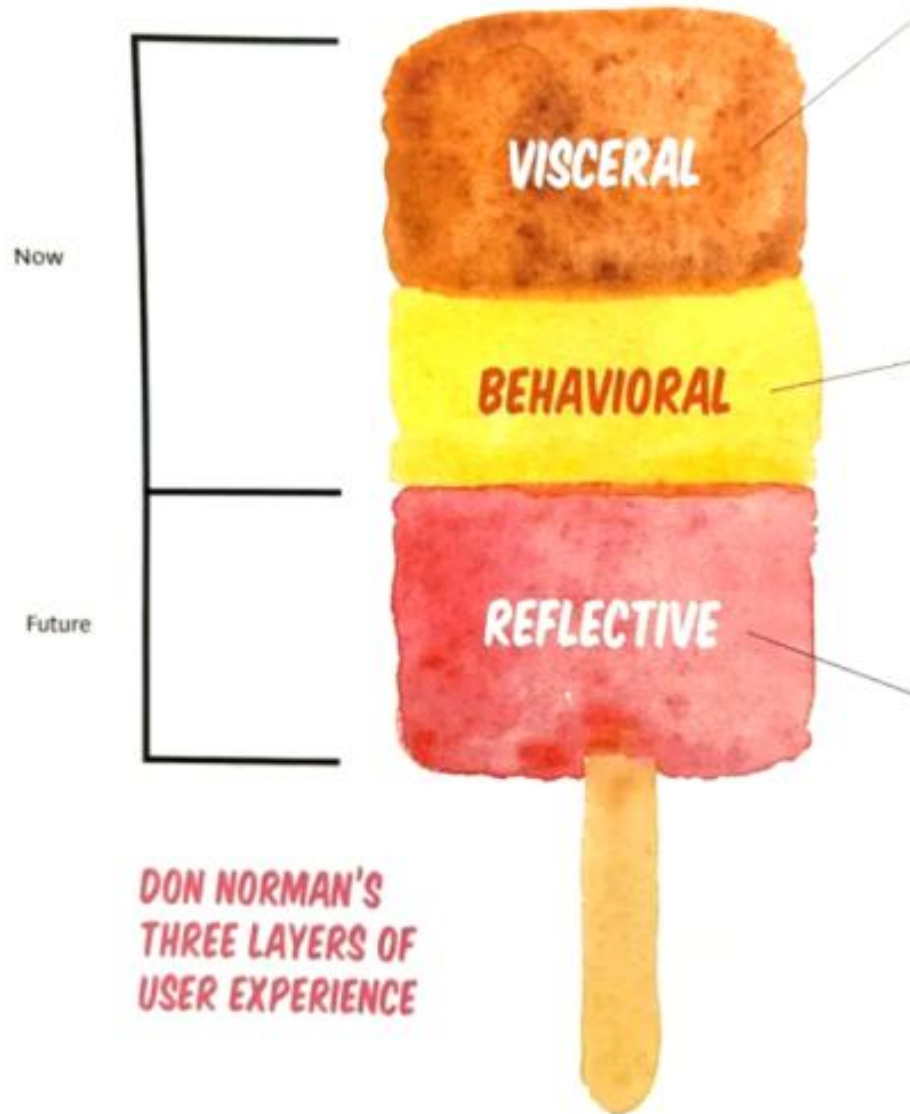
Experience Economy

The emotion wheel (Dr. Robert Plutchik's Wheel of Emotions) is a tool that helps to identify & label user's feelings by categorizing emotions into core, nuanced variations to facilitate better communication.



Experience Economy

This is how differently users respond to the user experience (or a story, a car ride, or anything else) with the passage of time (from the Beginning to the Future).



Beginning

At the visceral level, design stirs an immediate reaction to form, colour, texture and materials used.

Middle

At the behavioural level, design prompts a physical response or action.

End

At the reflective level, design engages memories and associations.

Experience Economy

*Adapted
Maslow's
Hierarchy
[of user
experience]*

Understanding user experience allows the designer to move up the hierarchy (i.e. from Functional to Pleasurable).

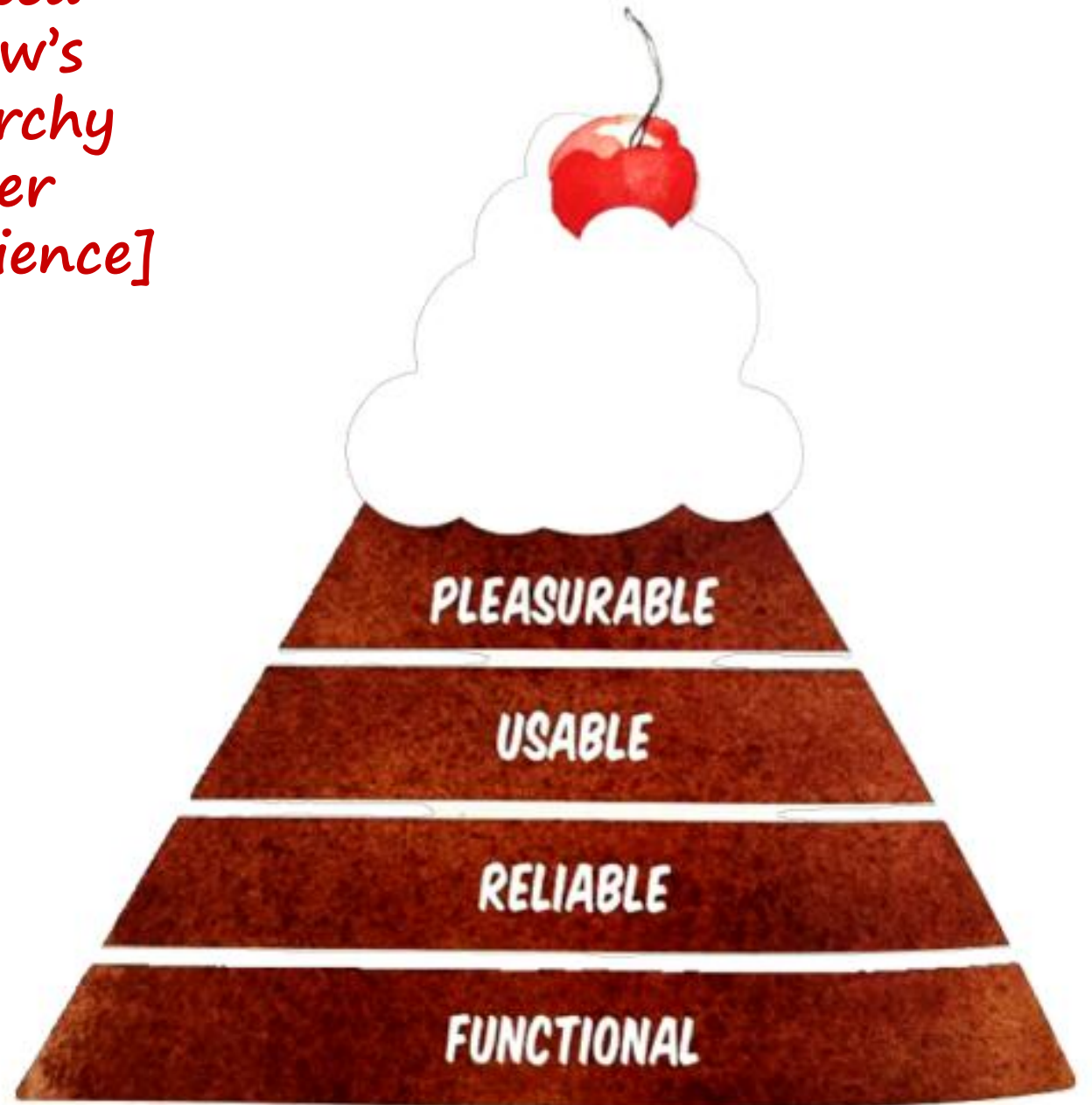
Functional: Does it work? If not, nothing else matters.

Reliable: Can users trust it to work consistently?

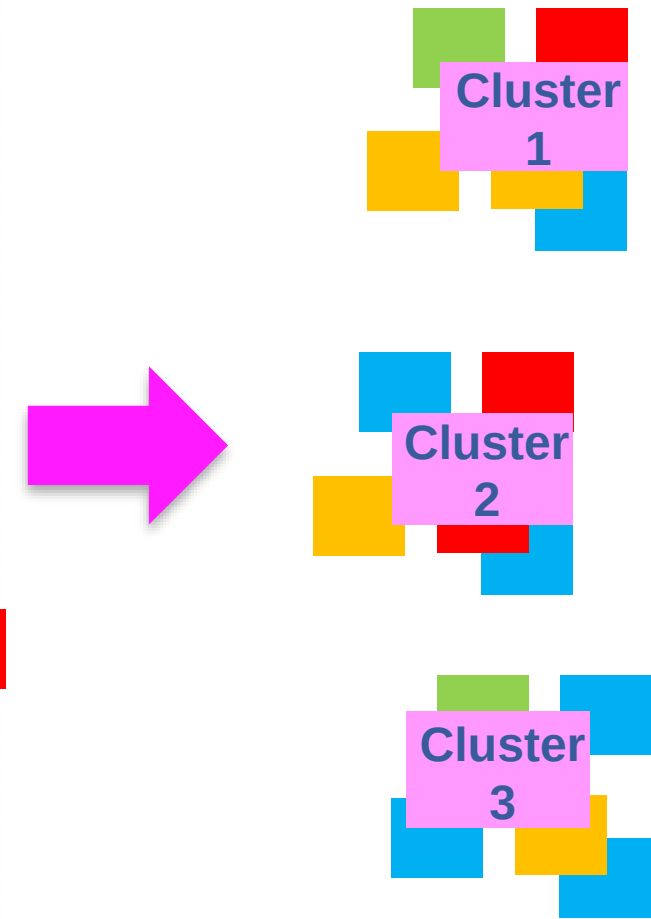
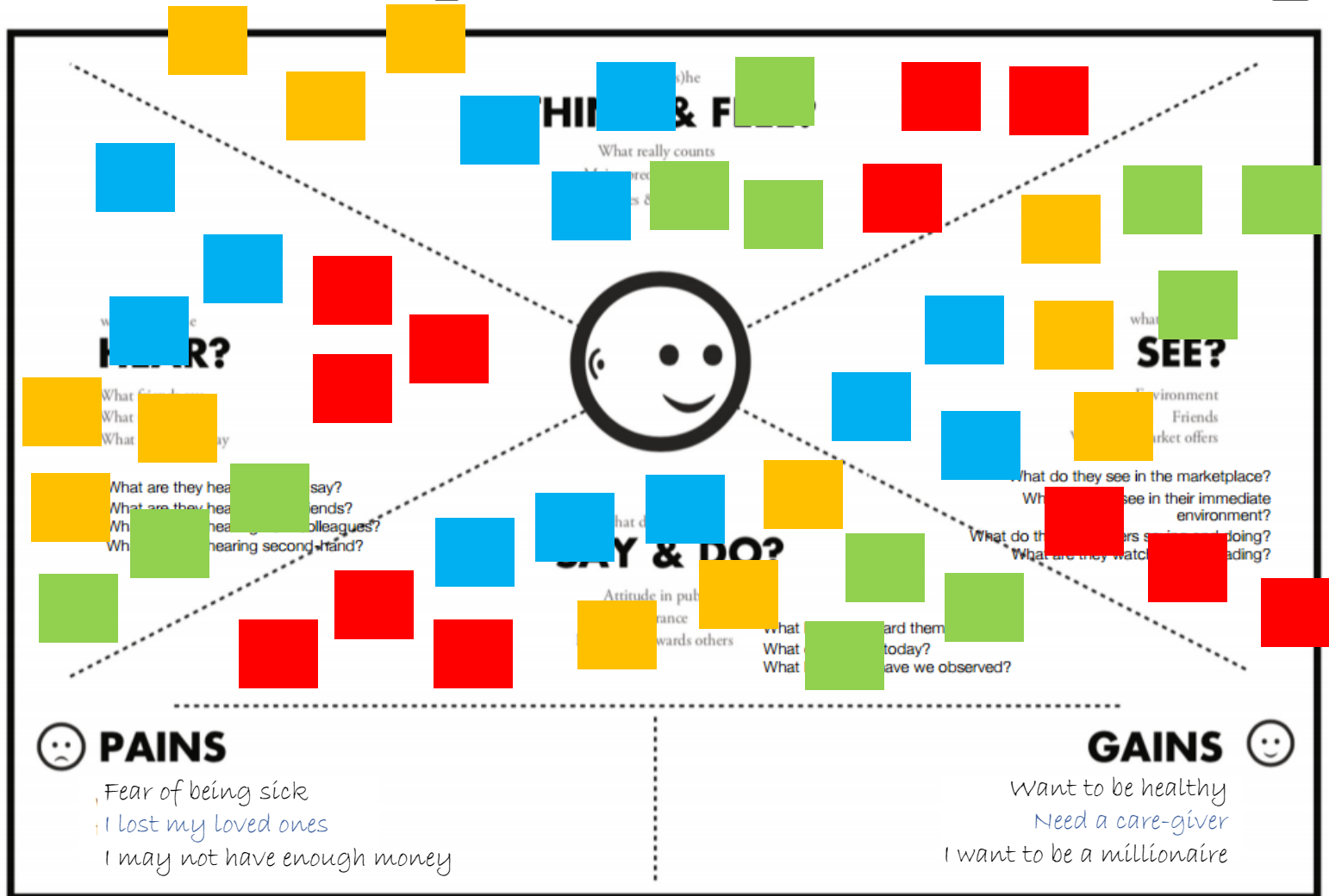
Usable: Is it easy and intuitive to use?

Pleasurable: Does it create moments of joy that users remember?

User delight is the cherry on top



Affinity Clustering





User Goals
Product Values
Product Features



User Goals – Why's behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.

Product Features – How the product delivers

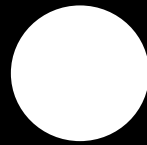
The specific functionalities or capabilities of a product.

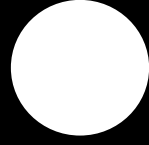
Product Values – What the product delivers

The desired outcomes or objectives that users aim to achieve by using a product or service.

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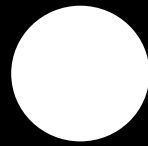


Product Values – What the product delivers

The desired outcomes or objectives that users aim to achieve by using a product or service.

Product Features – How the product delivers

The specific functionalities or capabilities of a product.



User Goals – Why's behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.

User Goals (Why?)

A tech savvy user wants to communicate with friends and family easily.



Product Values – What the product delivers

The core benefits that a product brings to its users.

User Goals (Why?)

A tech savvy user wants to communicate with friends and family easily.

Product Values (What?)

A means for the user to communicate with friends and family using her mobile devices



Product Features – How the product delivers

The specific functionalities or capabilities of a product.

User Goal (Why?)

A tech savvy user wants to communicate with friends and family easily.

Product Value (What?)

A means for the user to communicate with friends and family using her mobile devices

Product Feature (How?)

A mobile app that allows the user to post pictures and also send/receive messages on-the-go



User Goals – Why's behind the JTBD

A student wants to learn a new hobby at her own free time.



User Goal (Why?)

A student wants to learn a new hobby at her own free time.

Product Value (What?)

A means for her to learn about the new hobby at her own pace.

Product Feature (How?)

A online learning platform that tracks her learning progress.



Creative Sparks: Finding Ideas in the Unexpected



Innovation

A favourite word for both designers and engineers the development of new ideas or the improvement of existing ideas

but how?

How to be innovative?

Research? Ai?

Advancement in technology

Creativity

“Thinking” - How to think a new idea?



Activities

- Design Oxymoron Method
- The Crazy 8
- Crazy 8 Ideation + **Bouncing**
- Mid terms grading component

Activity Time!

~5 mins

A4 paper and writing materials!



Write down the key product you are going to innovate with E.g - Water bottle/Medicine dispenser/coaster

If your group does not have one, pick an item from below

Pen
Toothbrush
Keys

Exploring Design Oxymoron

What is a “Design Oxymoron”?

A combination of contradictory elements that provoke tradition and spark innovation.

Bladeless fan

Wireless charging

Air-less tyres



OXYMORON

(ok-see-MOR-on)

Definition:

An oxymoron pairs opposing terms, like "jumbo shrimp," to create intriguing language.

Ex Examples.com



Jumbo Shrimp



Deafening Silence



Virtual Reality



Living Dead

Functional roots

A statement that describes the basic function of a product (It's core function)

Fan *An object that displaces air with electricity that powers a moving surface*

Fan *An object that displaces air*

What is Oxymoron?

How does this spark thinking?



What is Oxymoron?

How does this  thinking?

Breaking down objects into their fundamental functions

Conventional

A fan that can blow more air = More blades?

A bigger fan for faster air flow = Bigger blades?

Unconventional (Oxymoron)

*Fan - An object that **displaces air***



What other ways can we displace air?

Now it gets interesting right?



Examples



Violin

Functional root?



Violin

Functional root : **Produces sound** by drawing a
bow across a string



Violin (Oxymoron)

A stringless violin? To reduce injuries
caused by the string



Candle

A source of fragrance and light
via a naked flame



Candle (Oxymoron)

Non-flammable Candles to reduce fire hazards associated to an open flame

Odor candles to repel people/insects away from a space



Wrist watch

A portable timekeeper worn on the wrist,
displaying time on it's front face



Wrist watch (oxymoron)

Faceless Wrist Watch to improve the timekeeping experience for the visually impaired

What makes a good Design Oxymoron statement?

- Contains a contradictory element
- Conveys a clear meaningful intent/objective
- Provides context how it's objective can be achieved.

E.g - Bladeless fans to improve user safety when operating with households with children

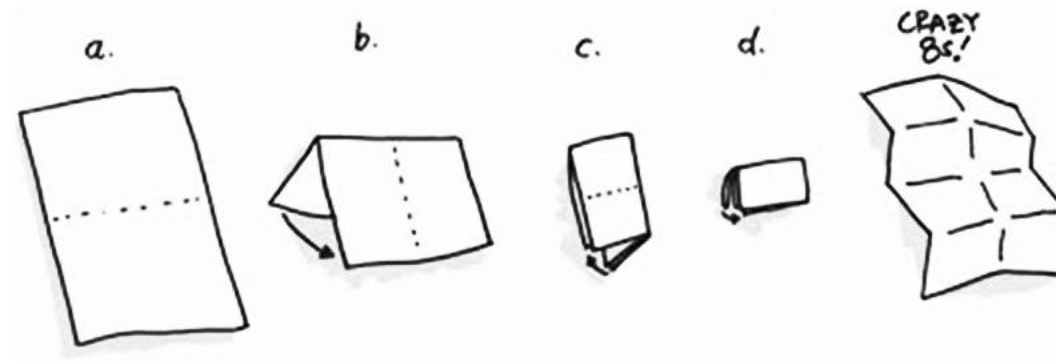


Write their functional roots & Oxymoron statement
~10 mins

Draw and annotate!

- Contains a contradictory element
- Conveys a clear meaningful intent/objective
- Provides context how it's objective can be achieved.

The Crazy 8 Technique



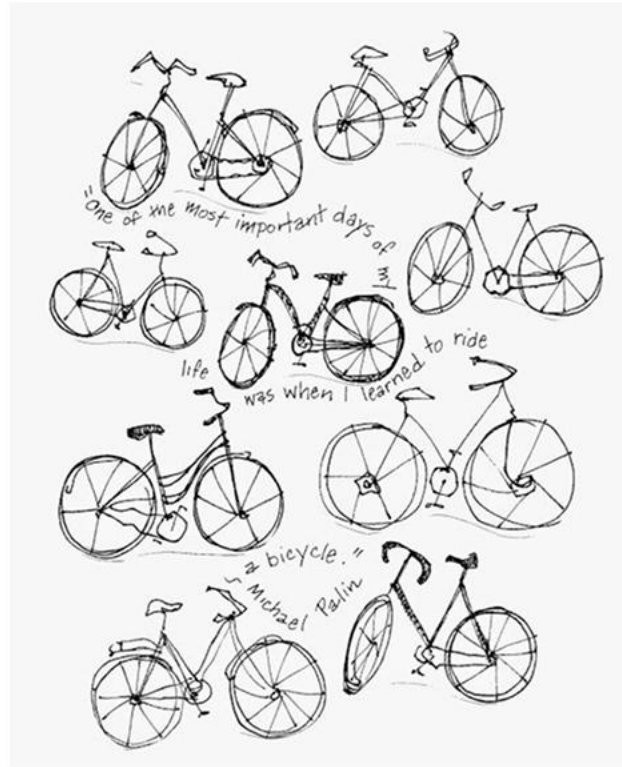
Just write, don't think

"Crazy 8 is a **fast-paced** exercise that forces you to **push past** your initial ideas and dive into **creative problem-solving**. By sketching eight ideas in just eight minutes, you tap into your brain's ability to **think laterally** and explore a range of solutions."

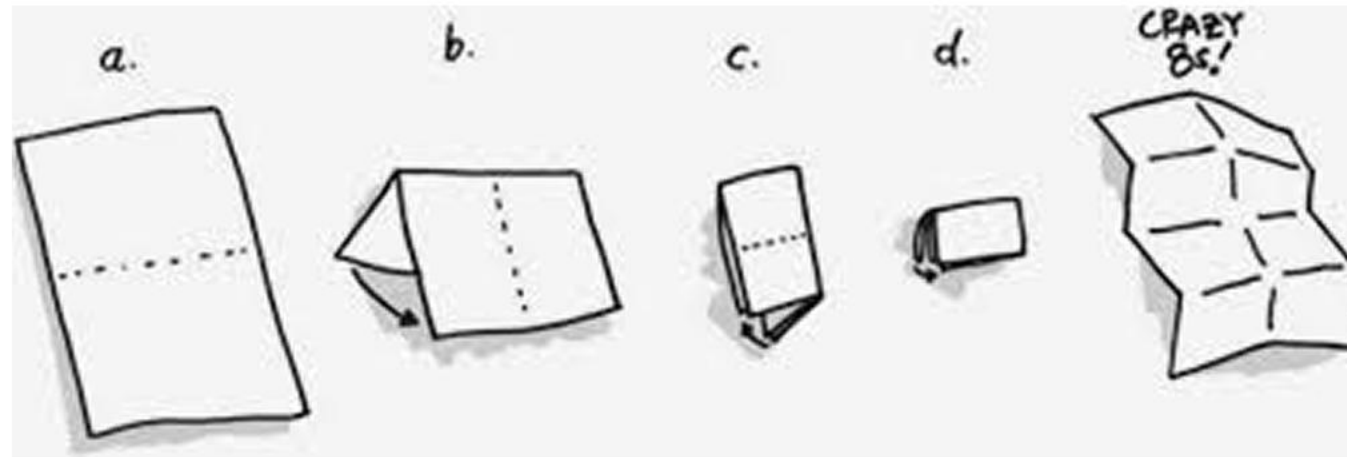
– Jake Knapp, *Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days*



Quantity vs Quality



Quantity vs Quality ...pst, you are not trying to an artist!!



Crazy 8 Ideation

~10 mins

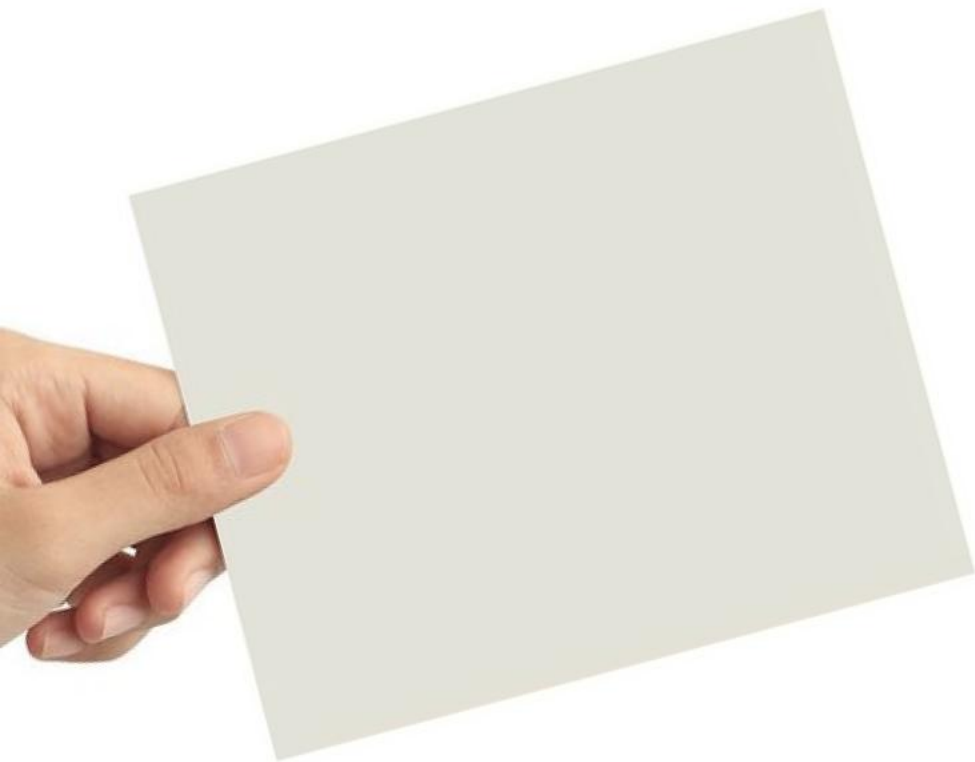
Think of 8 ways you can fulfil the functional root of the product

8mins total, 1mins each segment

Group Discussion
~10 mins

|

Each person, share your selected item, the design oxymoron statement and your concept



Crazy 8 + Bouncing



Bounce Phase
~8 mins



Pass your paper to another person, and now take 8mins to review the 8 ideas - Redraw with your own interpretation + suggestions

8mins total, 1mins each segment

**Reflection &
Group Discussion**
~10 mins

|

Everyone pass back the papers to the original owner and share your improvements with them

How is this useful to real projects?

Our case studies

Functional root (Scented candle)

A source of fragrance

A source of controlled fragrance

A source of fragrance made with
fragrant oils mixed into the wax, giving
off a pleasant aroma when burning

A source of controlled fragrance emitted
from burning fragrant oils

A source of controlled fragrance emitted
by diffusing fragrant oils

Initial Oxymoron drafts

Foul smelling candle as a defensive tool to deter potential
assailants from attacking/pursuing.

A weapon that does not injure or kill anyone, but instead
triggers fear, panic and an overwhelming urge to run
away.

(Self defensive weapon - Pepper spray)

Foul smelling candles to deter dieters from snacking

A vomit-scented "freshener" to deter dieters from snacking,
helping them stick to their diet.

Foul smelling candles to discourage smoking

Smelly cigarettes to deter behavior (Picking up the box)

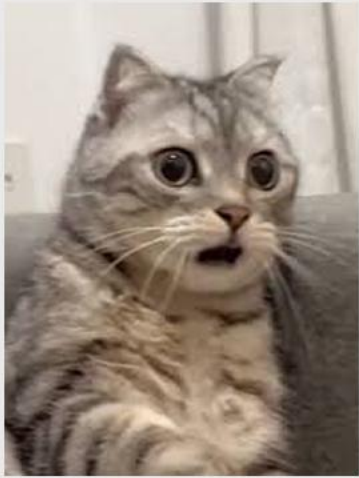
Foul smelling candles to educate and spread awareness
about the dangers associated to a foul smell.

Bad smell library to spread awareness and educate what a
foul smell means (Food spoil, gas leak, dangerous fumes)

Foul smelling candles to promote a sense of urgency
during a building evacuation.

Foul smell released during emergency to encourage
evacuation OR

Foul smells released to disperse crowd or vacate an area



**An Odor “Candle” to facilitate
indoor evacuations using odor
as a nudge**



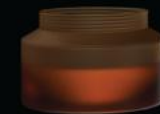
nudge

**A fire alarm
that works**

A new way to facilitate
evacuations during a
fire emergency, with
the power of odor

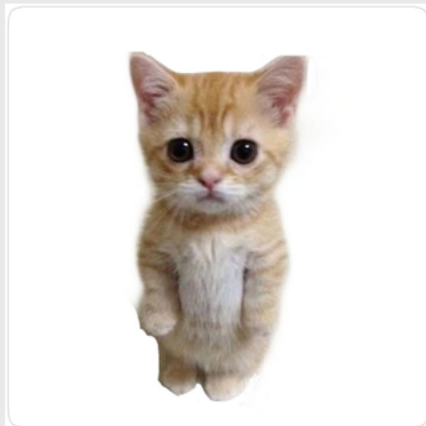


Odor dispenser



Odor refuel pod

Support why is this needed/useful



Why is this needed?

People becoming complacent and non-reactive to a real fire alarm, due to false alarms > Only panic once they smell or see smoke

When a fire alarm goes off people often feel uncertain about how they should respond. In a real fire any delay can be fatal.

Foul smell will motivate people to evacuate the area faster (Less hesitating and waiting around for a confirmation)

The sound of a fire alarm may be completely ignored, or occupants will look for the reaction of others before doing anything themselves. Panic can lead to distress and confusion, with occupants bypassing a fire exit in order to leave a building by the front door. 30 Jan 2015

<https://www.nature.com/articles/bdjteam20158>

How would you react in a fire? | BDJ Team - Nature

nudge is a new kind of fire alarm system that uses odor as a means to disrupt and nudge people out of a space facilitating faster evacuations during a fire emergency, **improving occupant survival rates by 66%***.

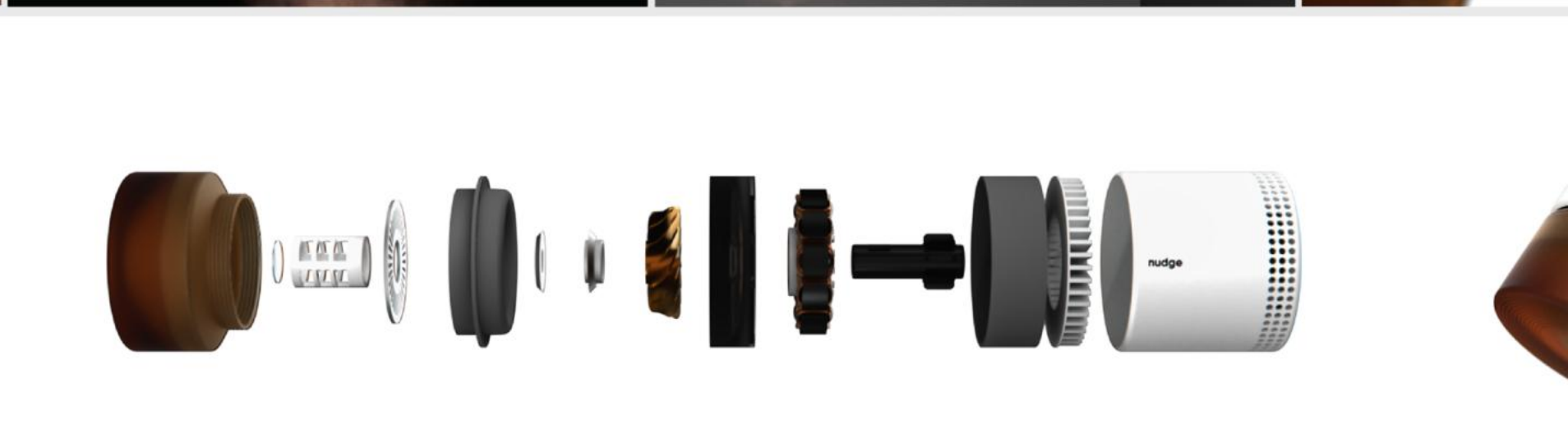
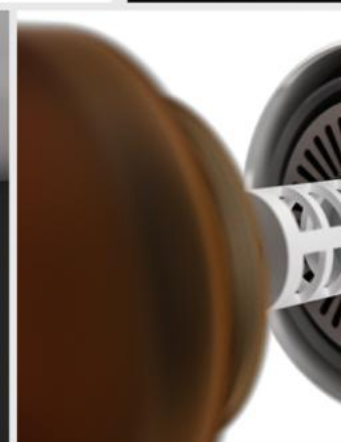
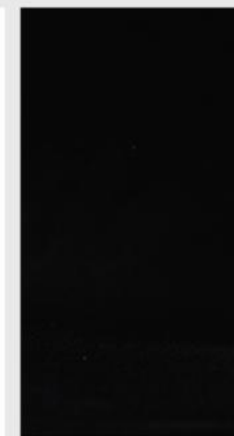
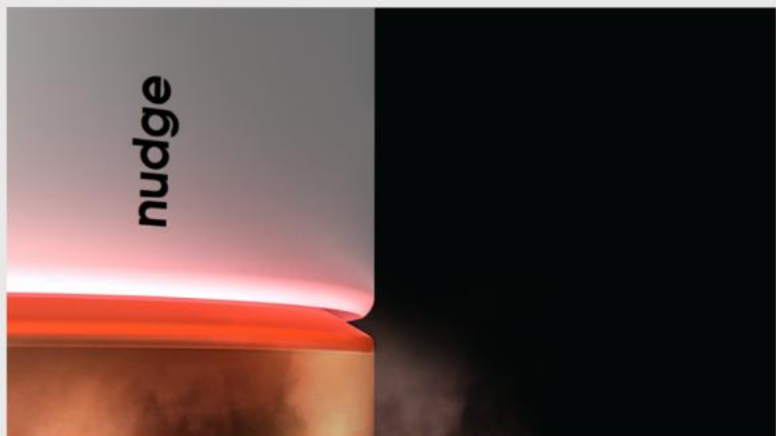
It taps on the existing fire detection systems to detect fires in the building. When a fire is detected, the formulated odor that smells like smoke is instantly diffused into the surroundings. It has been engineered to be compact yet containing an innovative ventilation system to rapidly throw the odor into the space.

The formulated smokey odor is safe to smell, dissipating quickly when left standing in a ventilated space.



* Based on the egress time requirements according to EU fire safety regulations, 2005. Test consists of 18 participants.

nudge | An odor candle used to nudge people, facilitating faster evacuations during a fire emergency





Mid terms briefing CDE 2300

12% of grade
6 mins per group
random order

Summary of Fieldwork and Affinity Clusters (25 marks)

- Clear explanation of problem/scenario identified
- Good **insights** identified from observation or research
- Clear indication that both **quantitative** and **qualitative** research

TLDR - Clear data/research and overview of problem and scope

Assessment Rubrics: Mid Term Review

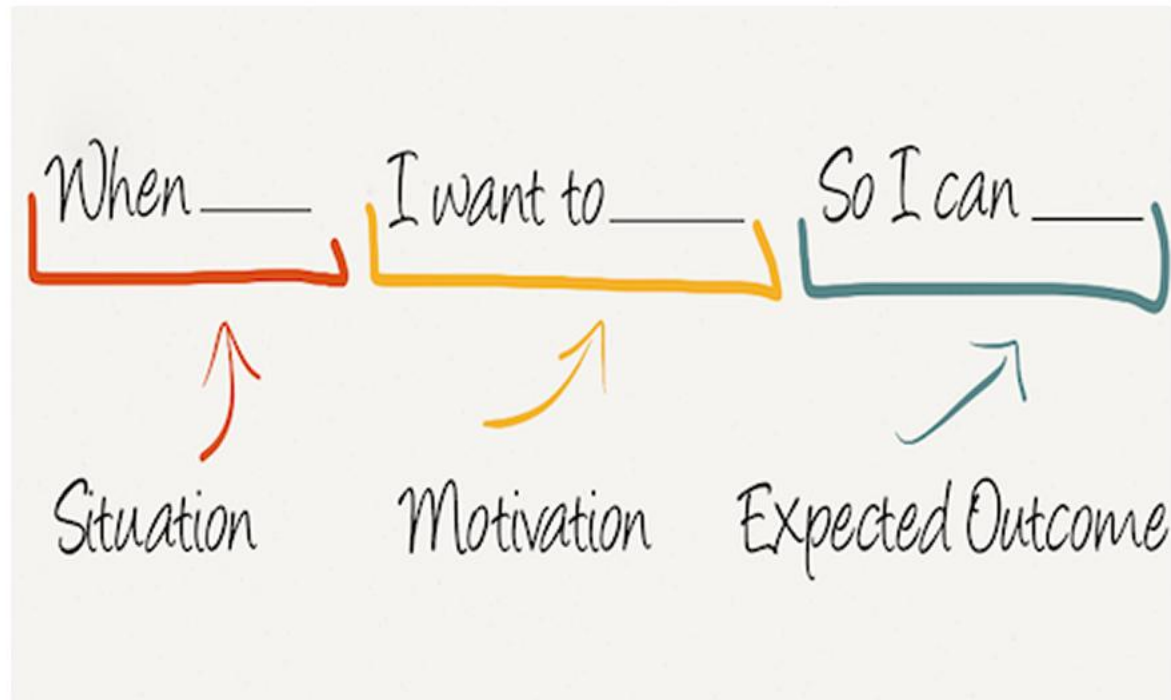
Competency	Accomplished	Capable	Apprentice	Unsatisfactory
1. Summary of Fieldwork and Affinity Clusters (25 marks)	Excellent Summary (score: 10-9) The main points in the summary provides a clear and concise overview of the scenario and users. Distinctive and interesting affinity clusters are identified.	Good Summary (score: 8-7) The points in the summary provides a reasonable overview of the scenario and users. Good affinity clusters are identified.	Adequate Summary (score: 6-5) The points in the summary provides a rough overview of the scenario and users. Some affinity clusters are identified.	Poor Summary (score: 4-0) The points in the summary provides a poor overview of the scenario. Affinity clusters are not identified.
2. Jobs-To-Be-Done (JTBD) (10 marks)	Excellent JTBD (score: 10-9) The JTBD covers functional, emotional, and social aspects accurately and comprehensively.	Good JTBD (score: 8-7) The JTBD covers functional, emotional, and social aspects accurately.	Adequate JTBD (score: 6-5) The JTBD covers functional, emotional, and social aspects.	Poor JTBD (score: 4-0) The JTBD has functional, emotional, and social aspects missing.
3. Persona(s) (20 marks)	Excellent Persona (score: 50-41) The description of the persona(s) is empathetic, singular, comprehensive and contextual. The relationships to the functional, emotional, and social aspects of the person(s) needs are accurately and comprehensively mapped out.	Good Persona (score: 40-21) The description of the persona(s) is empathetic, singular. Most of the main relationships to the functional, emotional, and social aspects of the person(s) needs are accurately mapped out.	Adequate Persona (score: 20-11) The description of the persona(s) is empathetic and singular. Some of the main relationships to the functional, emotional, and social aspects of the person(s) needs are mapped out, with some missing details.	Poor Persona (score: 10-0) The description of the persona(s) is not empathetic, singular, comprehensive and contextual. The main relationships to the functional, emotional, and social aspects of the person(s) needs are missing.
3. Product Value(s) (20 marks)	Excellent Product Value(s) (score: 10-9) The product value(s) are identified uniquely, accurately, and aligned with the needs of users.	Good Product Value(s) (score: 8-7) Most of the important product value(s) are identified accurately and aligned with the needs of users.	Adequate Product Value(s) (score: 6-5) Some of the product value(s) are identified with alignment with the needs of users.	Poor Product Value(s) (score: 4-0) The relevant product value(s) are not identified or aligned with the needs of users.
4. Product Vision	Excellent Product Vision (score: 10-9)	Good Product Vision (score: 8-7)	Adequate Product Vision (score: 6-5)	Poor Product Vision (score: 4-0)

Details will be shared

Jobs to be done (JTBD)
(10 marks)

- Functional, emotional, social expect
(Not every product needs to focus on all 3 -
focus on what matters)

**TLDR - Short, concise and easy to
understand/relate.**



Product value (10 marks)

- Product can perform to meet needs of your persona
- “What your customers get out of your product - and is it worth it?”
- Value is unique (bonus)

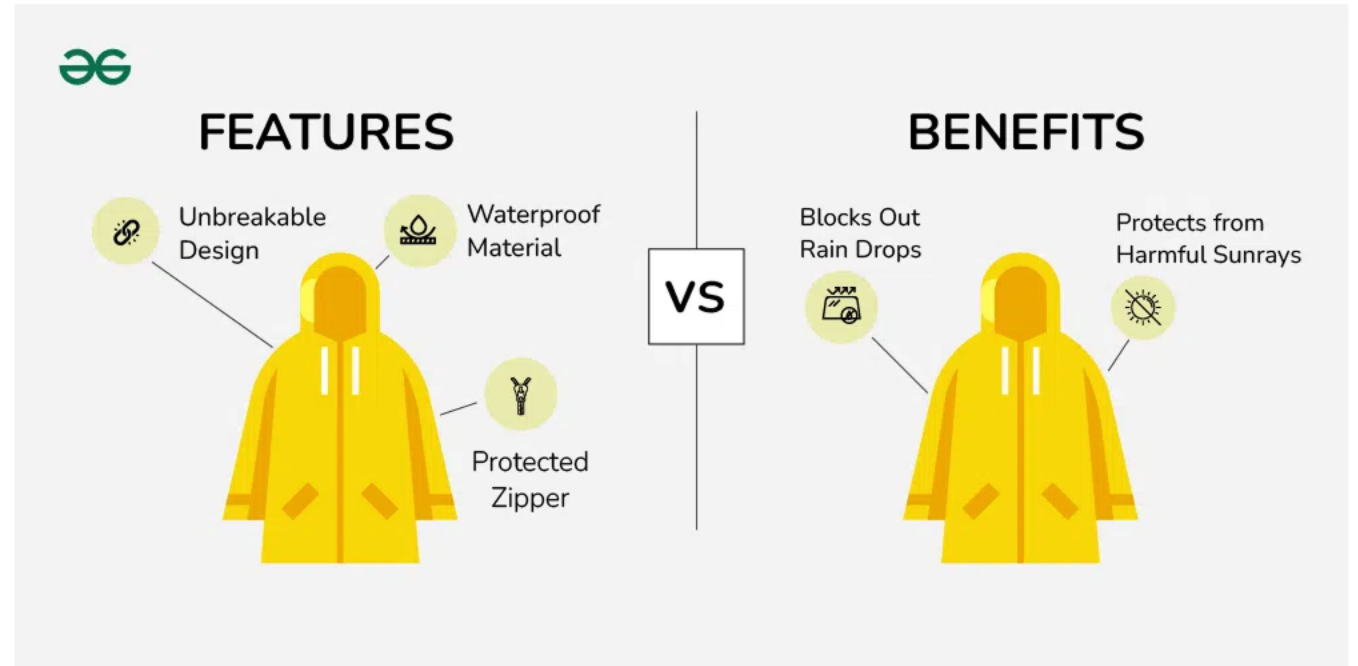
TLDR - Correctly identified what your user wants according to your persona's needs. Product provides benefits to your persona (bonus)



Product Features/Vision (20 marks)

- USPs
- Competitor analysis - why yours is/might be different or better than the best thing in the market
- Clear link of how the feature can help your product perform it's job

TLDR - Good USP stated, good analysis, clear link of how USP works for your persona



Presentation / Slides (5 marks)

- Clear visual hierarchy
- Readable fonts and good contrast

TLDR - Short, sweet and clear - no slide limit so make use of it

YOU At some point you may come back to read this line or maybe not.

**WILL READ
THIS FIRST.**

And then you will read this line next.

You will go back to read this body copy if you want to know more. It takes the most effort to read because it has a lot of text in a small font in a light weight with tight line spacing. Many people will skip paragraphs like this unless if they aren't engaged right away. This is why it's important to draw attention to your message using visual hierarchy.

You'll probably read this before the paragraph.

- **Design straight from Powerpoint '99**
- **Classic horrible font**
- **What are the key features?**
 - *Weird content nobody wants to learn*
 - *Lots of weird terminology to confuse*
 - *Clashing colours to make life difficult for those with visual difficulties*



Grim picture

Persona (20 marks)

- Correct persona identified for the problem scope
- persona is well defined
- persona has depth and **needs are well understood - relates to problem.**
- Every persona for every scope is presented differently

TLDR - persona must be relatable, containing details that relate to their needs or wants (not random information)

Kristin Watson



AGE	27
EDUCATION	Masters in Business
STATUS	Single
OCCUPATION	Sales Manager
LOCATION	Sydney
TECH LITERTE	High

“ I am used to with online service and I usually do my online shopping from Instagram

Personality

Introvert Thinker

Bio

She currently lives in Sydney. She finished her master in businesss and has just been promoted to Sales Manager. She is currently single and like to go out with friends on long holidays.

Core needs

- Need to find people with similar skills that can help her tackle company goals.
- View all her hirings in an overview
- The price of the service is very important

Frustrations

- Price is high related to quality they provide
- Currently finds perect people from past work relations, family, friends and within my circle and online which is tedious
- Not much choice and comparison not available

Brands

NASA HUAWEI facebook YouTube Instagram BELLE

Payment medium

Cash/Cheque Digital Payment

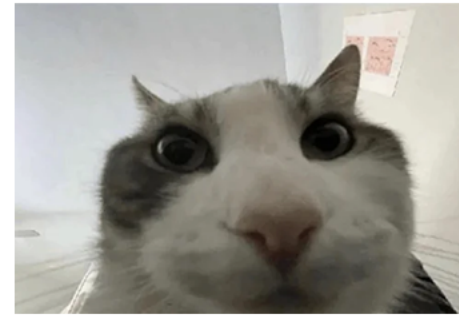
Platform

Website Mobile App

Tips

- Make your examples relatable / tell a story
- Sell your idea with only the key features (Top 3) , go deep into 3, rather than skipping over 6
- Ensure that the persona needs are realistic - Don't create your persona to have odd needs to match your product requirements forcefully

"I want a clear plastic bag that has compartments to store a certain box of food container (that i have designed)"



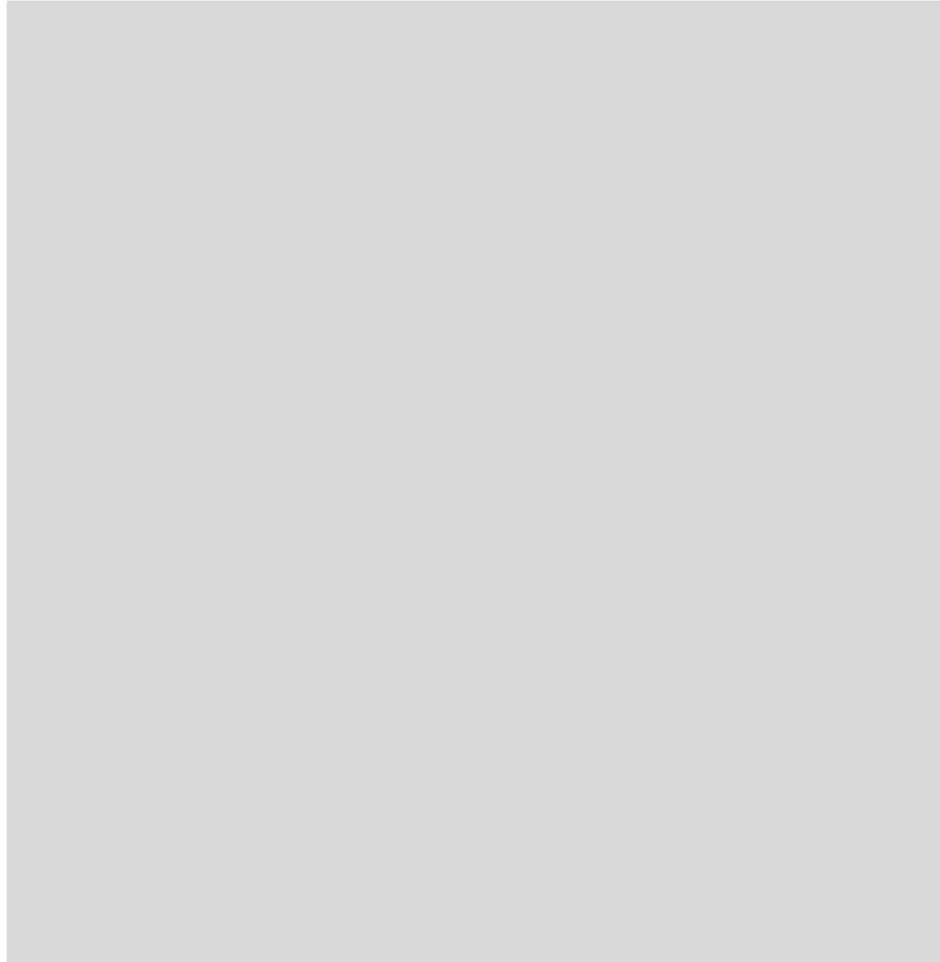
- Identify a problem and tell us why it matters. (problem/solution should be somewhat impactful)

"I want to keep my dog happy because it makes me happy"
- Keep to the time, we won't let you finish beyond 6mins
- Show that you did research beyond the internet (interviews, ideations)

Jobs to be done statement

Summary on the last slide

Persona



Product vision and feature



When I am inside a building and a fire emergency occurs, I want to be alerted in a way that feels instinctive, so I can react quickly.

Persona

Name - Alex

Role - Office Worker

Behaviour - Often wears noise-cancelling headphones and hyper-focused at work, and accustomed to “false alarm” drills

Pain Point - Traditional ringing alarms are often perceived as annoyances or drills rather than immediate threats, leading to delayed reaction times. Alex needs a signal that bypasses “interpreting” noise and triggers a primal flight response.

Product vision and feature

Vision

- To revolutionise building safety by harnessing the primal power of olfactory processing - creating a fire alarm system that triggers an immediate biological instinct of danger, thereby reducing reaction time and improving survival rates.

Features

- Olfactory alert mechanism - releases an odor that mimics the smell of smoke to trigger an instinctive danger response.

-Rapid diffusion - ventilation system designed to throw the small rapidly across the room to ensure immediate coverage.

-Retrofit integration - designed to work alongside existing fire detection systems rather than replacing them